

Juan David Muñoz-Bolaños

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Carl Zeiss Vision GmbH

Attn: Hiring Team
Gartenstraße 97
73430 Aalen, Germany

Subject: Application for the Expert Vision Science Position

Dear Ms. Beutler,

Following my participation in the ****ZEISS Summer School 2025****, I am motivated to bring my expertise in biomedical optics to the Vision Care team in Aalen. As a Biomedical Optical Engineer concluding my PhD at the Medical University of Innsbruck. I am writing to apply for the Expert Vision Science position. My background in adaptive optics/microscopy and vision aligns with the technical requirements of your Aalen-based team.

During my PhD research in Innsbruck, I used adaptive optics to correct optical aberrations from the microscope and biological tissue, this is similar to the human eye aberrations. In my master in Jena, I imaged cancer tissue to identify malignant tissue by using machine learning algorithms. I also explored the industry in Medellín dealing with computation and machine vision systems.

I have the following skills that I believe would be valuable to your team:

Practical Vision Science & Solving problems:

Experience designing and implementing optical microscopes for improving medical imaging. My work has studied cancer identification by machine learning algorithms and mouse brain imaging using adaptive optics.

Optical Simulation & Tolerance Analysis:

I can offer a bridge between physical-based modeling and Ophthalmic reality. My experience in simulating computationally complex bio-optical systems allows me to foresee biological and optical interactions.

Physiological Optics

My research in cancer imaging in Leibniz IPHT Jena and adaptive optics for image restoration in Innsbruck show me the foundation necessary to understand and optimize human vision, sharpness, binocular, comfort.

I am Colombian national with a German master's degree in Photonics, and soon PhD in Biomedical Optics with good B2 German certificated skills. This makes me qualify as graduated from Austrian/German university which makes hiring process seamless. Additionally, I have the willingness to relocate to Aalen.

I am available for an interview at your convenience and look forward to chatting how my expertise in vision science and biomedical optics expertise can contribute to ZEISS's. My earliest possible start date is from April/May 2026. Thank you for considering my application.

Sincerely,

Juan David Muñoz-Bolaños

Biomedical Optical Engineer
Medical University of Innsbruck